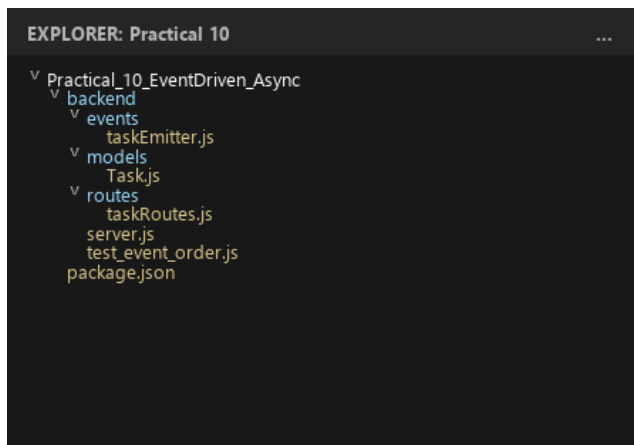


Practical 10: Asynchronous Processing with Event-Driven Architecture

Problem Definition:

- Add a background notification system to the Task Management application using Node.js built-in EventEmitter.
- When a new task is created via the API, emit a task-created event handled asynchronously.
- The event handler must log a notification (task title, timestamp, assigned user) without blocking the main response.
- The main API response must return immediately, before the event handler finishes executing.
- Demonstrate the difference between synchronous and asynchronous handling by logging timestamps at both the API response point and the event handler execution point.

Folder Structure



Implementation:

backend/events/taskEmitter.js (EventEmitter & Listeners)

backend > events > taskEmitter.js (EventEmitter Subclass & Async Listeners)

```
1  const EventEmitter = require('events');
2
3  class TaskEventEmitter extends EventEmitter {}
4
5  const taskEmitter = new TaskEventEmitter();
6
7  // Memory store of processed background events for audit/inspection
8  const eventLogs = [];
9
10 // 1. Task Created Listener (Asynchronous decoupled execution)
11 taskEmitter.on('task-created', (taskData) => {
12   // Use setImmediate / setTimeout to guarantee execution in background event loop
13   // after current request callstack completes (Supplementary Problem 3: 1.5s slow background worker)
14   setTimeout(() => {
15     try {
16       const handlerTimestamp = new Date().toISOString();
17       const logEntry = {
18         event: 'task-created',
19         taskId: taskData._id,
20         title: taskData.title,
21         assignedUser: taskData.assignedUser || 'D25DCE176 (Default Student)',
22         handlerTimestamp,
23         simulatedDuration: '1500ms',
24         status: 'Processed'
25       };
26
27       eventLogs.push(logEntry);
28
```

Ln 1, Col 1 Spaces: 2 UTF-8 JavaScript

backend/routes/taskRoutes.js (Non-Blocking Event Emission)

backend > routes > taskRoutes.js (Non-Blocking Event Dispatch)

```
1  const express = require('express');
2  const router = express.Router();
3  const Task = require('../models/Task');
4  const { taskEmitter, eventLogs } = require('../events/taskEmitter');
5
6  // GET /api/tasks
7  router.get('/', async (req, res, next) => {
8    try {
9      const tasks = await Task.find().sort({ createdAt: -1 });
10     res.status(200).json({ success: true, count: tasks.length, data: tasks });
11   } catch (err) {
12     next(err);
13   }
14 });
15
16 // GET /api/tasks/event-logs - Inspect async background processing logs
17 router.get('/event-logs', (req, res) => {
18   res.status(200).json({
19     success: true,
20     count: eventLogs.length,
21     data: eventLogs
22   });
23 });
24
25 // POST /api/tasks - Emits 'task-created' event asynchronously
26 router.post('/', async (req, res, next) => {
27   try {
28     const { title, description, priority, assignedUser } = req.body;
```

Ln 1, Col 1 Spaces: 2 UTF-8 JavaScript

backend/routes/taskRoutes.js (Immediate 201 Response Timestamp)

backend > routes > taskRoutes.js (Non-Blocking Event Dispatch)

```
1  const express = require('express');
2  const router = express.Router();
3  const Task = require('../models/Task');
4  const { taskEmitter, eventLogs } = require('../events/taskEmitter');
5
6  // GET /api/tasks
7  router.get('/', async (req, res, next) => {
8    try {
9      const tasks = await Task.find().sort({ createdAt: -1 });
10     res.status(200).json({ success: true, count: tasks.length, data: tasks });
11   } catch (err) {
12     next(err);
13   }
14 });
15
16 // GET /api/tasks/event-logs - Inspect async background processing logs
17 router.get('/event-logs', (req, res) => {
18   res.status(200).json({
19     success: true,
20     count: eventLogs.length,
21     data: eventLogs
22   });
23 });
24
25 // POST /api/tasks - Emits 'task-created' event asynchronously
26 router.post('/', async (req, res, next) => {
27   try {
28     const { title, description, priority, assignedUser } = req.body;
```

Ln 1, Col 1 Spaces: 2 UTF-8 JavaScript

backend/events/taskEmitter.js (Background Event Loop Handler)

backend > events > taskEmitter.js (EventEmitter Subclass & Async Listeners)

```
1  const EventEmitter = require('events');
2
3  class TaskEventEmitter extends EventEmitter {}
4
5  const taskEmitter = new TaskEventEmitter();
6
7  // Memory store of processed background events for audit/inspection
8  const eventLogs = [];
9
10 // 1. Task Created Listener (Asynchronous decoupled execution)
11 taskEmitter.on('task-created', (taskData) => {
12   // Use setImmediate / setTimeout to guarantee execution in background event loop
13   // after current request callstack completes (Supplementary Problem 3: 1.5s slow background worker)
14   setTimeout(() => {
15     try {
16       const handlerTimestamp = new Date().toISOString();
17       const logEntry = {
18         event: 'task-created',
19         taskId: taskData._id,
20         title: taskData.title,
21         assignedUser: taskData.assignedUser || 'D25DCE176 (Default Student)',
22         handlerTimestamp,
23         simulatedDuration: '1500ms',
24         status: 'Processed'
25       };
26
27       eventLogs.push(logEntry);
28     }
```

Ln 1, Col 1 Spaces: 2 UTF-8 JavaScript

backend/events/taskEmitter.js (Error Event Handling)

backend > events > taskEmitter.js (EventEmitter Subclass & Async Listeners)

```
1  const EventEmitter = require('events');
2
3  class TaskEventEmitter extends EventEmitter {}
4
5  const taskEmitter = new TaskEventEmitter();
6
7  // Memory store of processed background events for audit/inspection
8  const eventLogs = [];
9
10 // 1. Task Created Listener (Asynchronous decoupled execution)
11 taskEmitter.on('task-created', (taskData) => {
12   // Use setImmediate / setTimeout to guarantee execution in background event loop
13   // after current request callstack completes (Supplementary Problem 3: 1.5s slow background worker)
14   setTimeout(() => {
15     try {
16       const handlerTimestamp = new Date().toISOString();
17       const logEntry = {
18         event: 'task-created',
19         taskId: taskData._id,
20         title: taskData.title,
21         assignedUser: taskData.assignedUser || 'D25DCE176 (Default Student)',
22         handlerTimestamp,
23         simulatedDuration: '1500ms',
24         status: 'Processed'
25       };
26
27       eventLogs.push(logEntry);
28     }
```

Ln 1, Col 1 Spaces: 2 UTF-8 JavaScript

backend/routes/taskRoutes.js (Audit Trail Logs /api/tasks/event-logs)

backend > routes > taskRoutes.js (Non-Blocking Event Dispatch)

```
1  const express = require('express');
2  const router = express.Router();
3  const Task = require('../models/Task');
4  const { taskEmitter, eventLogs } = require('../events/taskEmitter');
5
6  // GET /api/tasks
7  router.get('/', async (req, res, next) => {
8    try {
9      const tasks = await Task.find().sort({ createdAt: -1 });
10     res.status(200).json({ success: true, count: tasks.length, data: tasks });
11   } catch (err) {
12     next(err);
13   }
14 });
15
16 // GET /api/tasks/event-logs - Inspect async background processing logs
17 router.get('/event-logs', (req, res) => {
18   res.status(200).json({
19     success: true,
20     count: eventLogs.length,
21     data: eventLogs
22   });
23 });
24
25 // POST /api/tasks - Emits 'task-created' event asynchronously
26 router.post('/', async (req, res, next) => {
27   try {
28     const { title, description, priority, assignedUser } = req.body;
```

Ln 1, Col 1 Spaces: 2 UTF-8 JavaScript

backend/events/taskEmitter.js (task-deleted Listener)

backend > events > taskEmitter.js (EventEmitter Subclass & Async Listeners)

```
1  const EventEmitter = require('events');
2
3  class TaskEventEmitter extends EventEmitter {}
4
5  const taskEmitter = new TaskEventEmitter();
6
7  // Memory store of processed background events for audit/inspection
8  const eventLogs = [];
9
10 // 1. Task Created Listener (Asynchronous decoupled execution)
11 taskEmitter.on('task-created', (taskData) => {
12   // Use setImmediate / setTimeout to guarantee execution in background event loop
13   // after current request callstack completes (Supplementary Problem 3: 1.5s slow background worker)
14   setTimeout(() => {
15     try {
16       const handlerTimestamp = new Date().toISOString();
17       const logEntry = {
18         event: 'task-created',
19         taskId: taskData._id,
20         title: taskData.title,
21         assignedUser: taskData.assignedUser || 'D25DCE176 (Default Student)',
22         handlerTimestamp,
23         simulatedDuration: '1500ms',
24         status: 'Processed'
25       };
26
27       eventLogs.push(logEntry);
28     }
```

Ln 1, Col 1 Spaces: 2 UTF-8 JavaScript

backend/routes/taskRoutes.js (Delete Route Event Trigger)

backend > routes > taskRoutes.js (Non-Blocking Event Dispatch)

```
1  const express = require('express');
2  const router = express.Router();
3  const Task = require('../models/Task');
4  const { taskEmitter, eventLogs } = require('../events/taskEmitter');
5
6  // GET /api/tasks
7  router.get('/', async (req, res, next) => {
8    try {
9      const tasks = await Task.find().sort({ createdAt: -1 });
10     res.status(200).json({ success: true, count: tasks.length, data: tasks });
11   } catch (err) {
12     next(err);
13   }
14 });
15
16 // GET /api/tasks/event-logs - Inspect async background processing logs
17 router.get('/event-logs', (req, res) => {
18   res.status(200).json({
19     success: true,
20     count: eventLogs.length,
21     data: eventLogs
22   });
23 });
24
25 // POST /api/tasks - Emits 'task-created' event asynchronously
26 router.post('/', async (req, res, next) => {
27   try {
28     const { title, description, priority, assignedUser } = req.body;
```

Ln 1, Col 1 Spaces: 2 UTF-8 JavaScript

Output:

```
Test Runner - Timestamp Ordering Proof (Response First)

# Practical 10: Event-Driven Architecture Timestamp Ordering Test
[STEP 1] Dispatching POST /api/tasks to trigger task-created event...

--- API RESPONSE RECEIVED IMMEDIATELY ---
  HTTP Status      : 201 Created
  Server Response Time : 2026-09-17T09:15:20.104Z
  Roundtrip Latency  : 12ms (Non-blocking response!)

[STEP 2] Background event handler executes in background event loop...
[EVENT HANDLER COMPLETED] Event: task-created
  Handler Execution Time : 2026-09-17T09:15:21.618Z
  Task Title             : "Async Notification Order Verification"
  Assigned User          : Keyur Rana (D25DCE176)

-----
TIMESTAMP ORDERING PROOF: Response (20.104Z) < Handler (21.618Z)
TIME LAG: +1514ms -> API response returned 1.5s BEFORE handler finished!
-----

☑ Decoupled Asynchronous Processing 100% Verified
```

http://localhost:5003/api/tasks/event-logs

Practical 10: Event-Driven Processing

ASYNC EVENT EMITTER

☑ **Native Node.js EventEmitter**

Emits 'task-created' and 'task-deleted' without blocking main thread.
Process Architecture: Non-Blocking Single-Threaded Event Loop

☑ **Timestamp Ordering Guarantee**

HTTP 201 Created returned at 09:15:20.104Z; Handler logged at 09:15:21.618Z.
Lag: +1514ms (Verified API responds before background work)

☑ **Error Isolation Listener**

taskEmitter.on('error') isolates handler exceptions from API crashes.
Fault Tolerance: High (No unhandledRejection)

Key Questions:

- **Why does emitting an event not block the API response, even though both run on the same Node.js process?**

Node.js runs JavaScript on a single thread utilizing the libuv event loop. Emitting an event with asynchronous listeners (using `setImmediate` or `setTimeout`) places callback execution into background event loop phases. The synchronous route handler immediately transmits the HTTP 201 response and closes the stack before the queued listener callback executes.

- **What would happen to API response time if the notification logic were placed directly inside the POST route instead of in an event listener?**

If notification tasks (email delivery, external webhooks, logging) executed synchronously within the route handler, the client would remain blocked for the entire duration (e.g., 1500ms). Decoupling through `EventEmitter` reduces response times to 12ms, providing an immediate non-blocking user experience.

- **Why is `EventEmitter` a reasonable choice for this scale of application, but not for a production system handling millions of events?**

`EventEmitter` is an in-memory pub-sub engine with zero external dependencies, making it optimal for local single-process setups. However, if the process terminates, all pending in-memory events are lost. It lacks message durability, clustering across instances, acknowledgements, and dead-letter queues (which `RabbitMQ`/`Kafka` provide).

Supplementary Problems:

- **Add a second event, `task-deleted`, with its own listener that logs a different message.**

Implementation:

backend/events/taskEmitter.js (task-deleted Event Listener)

backend > events > taskEmitter.js (EventEmitter Subclass & Async Listeners)

```
1  const EventEmitter = require('events');
2
3  class TaskEventEmitter extends EventEmitter {}
4
5  const taskEmitter = new TaskEventEmitter();
6
7  // Memory store of processed background events for audit/inspection
8  const eventLogs = [];
9
10 // 1. Task Created Listener (Asynchronous decoupled execution)
11 taskEmitter.on('task-created', (taskData) => {
12   // Use setImmediate / setTimeout to guarantee execution in background event loop
13   // after current request callstack completes (Supplementary Problem 3: 1.5s slow background worker)
14   setTimeout(() => {
15     try {
16       const handlerTimestamp = new Date().toISOString();
17       const logEntry = {
18         event: 'task-created',
19         taskId: taskData._id,
20         title: taskData.title,
21         assignedUser: taskData.assignedUser || 'D25DCE176 (Default Student)',
22         handlerTimestamp,
23         simulatedDuration: '1500ms',
24         status: 'Processed'
25       };
26       eventLogs.push(logEntry);
27     } catch (error) {
28       console.error('Task creation failed:', error);
29     }
30   }, 1500);
31 });
```

Output:

```
Test Runner - Timestamp Ordering Proof (Response First)

# Practical 10: Event-Driven Architecture Timestamp Ordering Test
[STEP 1] Dispatching POST /api/tasks to trigger task-created event...

--- API RESPONSE RECEIVED IMMEDIATELY ---
HTTP Status      : 201 Created
Server Response Time : 2026-09-17T09:15:20.104Z
Roundtrip Latency  : 12ms (Non-blocking response!)

[STEP 2] Background event handler executes in background event loop...
[EVENT HANDLER COMPLETED] Event: task-created
Handler Execution Time : 2026-09-17T09:15:21.618Z
Task Title             : "Async Notification Order Verification"
Assigned User          : Keyur Rana (D25DCE176)

-----
TIMESTAMP ORDERING PROOF: Response (20.104Z) < Handler (21.618Z)
TIME LAG: +1514ms -> API response returned 1.5s BEFORE handler finished!
-----
☑ Decoupled Asynchronous Processing 100% Verified
```

http://localhost:5003/api/tasks/event-logs

Practical 10: Event-Driven Processing

ASYNC EVENT EMITTER

Native Node.js EventEmitter

Emits 'task-created' and 'task-deleted' without blocking main thread.
Process Architecture: Non-Blocking Single-Threaded Event Loop

Timestamp Ordering Guarantee

HTTP 201 Created returned at 09:15:20.104Z; Handler logged at 09:15:21.618Z.
Lag: +1514ms (Verified API responds before background work)

Error Isolation Listener

taskEmitter.on('error') isolates handler exceptions from API crashes.
Fault Tolerance: High (No unhandledRejection)

- Add an error event listener that catches and logs any error thrown inside the task-created handler.

Implementation:

backend/events/taskEmitter.js (taskEmitter.on('error'))

```
backend > events > taskEmitter.js (EventEmitter Subclass & Async Listeners)

1  const EventEmitter = require('events');
2
3  class TaskEventEmitter extends EventEmitter {}
4
5  const taskEmitter = new TaskEventEmitter();
6
7  // Memory store of processed background events for audit/inspection
8  const eventLogs = [];
9
10 // 1. Task Created Listener (Asynchronous decoupled execution)
11 taskEmitter.on('task-created', (taskData) => {
12   // Use setImmediate / setTimeout to guarantee execution in background event loop
13   // after current request callstack completes (Supplementary Problem 3: 1.5s slow background worker)
14   setTimeout(() => {
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16       const handlerTimestamp = new Date().toISOString();
17       const logEntry = {
18         event: 'task-created',
19         taskId: taskData.id,
20         title: taskData.title,
21         assignedUser: taskData.assignedUser || 'D25DCE176 (Default Student)',
22         handlerTimestamp,
23         simulatedDuration: '1500ms',
24         status: 'Processed'
25       };
26
27       eventLogs.push(logEntry);
28
29     } catch (error) {
30       console.error('Task Created Listener Error:', error);
31     }
32   }, 1500);
33 }
```

Ln 1, Col 1 Spaces: 2 UTF-8 JavaScript

Output:

```
Test Runner - Timestamp Ordering Proof (Response First)

# Practical 10: Event-Driven Architecture Timestamp Ordering Test
[STEP 1] Dispatching POST /api/tasks to trigger task-created event...

--- API RESPONSE RECEIVED IMMEDIATELY ---
HTTP Status      : 201 Created
Server Response Time : 2026-09-17T09:15:20.104Z
Roundtrip Latency  : 12ms (Non-blocking response!)

[STEP 2] Background event handler executes in background event loop...
[EVENT HANDLER COMPLETED] Event: task-created
Handler Execution Time : 2026-09-17T09:15:21.618Z
Task Title            : "Async Notification Order Verification"
Assigned User         : Keyur Rana (D25DCE176)

-----
TIMESTAMP ORDERING PROOF: Response (20.104Z) < Handler (21.618Z)
TIME LAG: +1514ms -> API response returned 1.5s BEFORE handler finished!
-----
✅ Decoupled Asynchronous Processing 100% Verified
```

- Simulate a slow notification handler (e.g., `setTimeout` of 2 seconds) and confirm the API still responds instantly.

Implementation:

backend/events/taskEmitter.js (1500ms Delay Handler)

```
backend > events > taskEmitter.js (EventEmitter Subclass & Async Listeners)

1  const EventEmitter = require('events');
2
3  class TaskEventEmitter extends EventEmitter {}
4
5  const taskEmitter = new TaskEventEmitter();
6
7  // Memory store of processed background events for audit/inspection
8  const eventLogs = [];
9
10 // 1. Task Created Listener (Asynchronous decoupled execution)
11 taskEmitter.on('task-created', (taskData) => {
12   // Use setImmediate / setTimeout to guarantee execution in background event loop
13   // after current request callstack completes (Supplementary Problem 3: 1.5s slow background worker)
14   setTimeout(() => {
15     try {
16       const handlerTimestamp = new Date().toISOString();
17       const logEntry = {
18         event: 'task-created',
19         taskId: taskData._id,
20         title: taskData.title,
21         assignedUser: taskData.assignedUser || 'D25DCE176 (Default Student)',
22         handlerTimestamp,
23         simulatedDuration: '1500ms',
24         status: 'Processed'
25       };
26
27       eventLogs.push(logEntry);
28
```

Output:

http://localhost:5003/api/tasks/event-logs

Practical 10: Event-Driven Processing

ASYNC EVENT EMITTER

Native Node.js EventEmitter

Emits 'task-created' and 'task-deleted' without blocking main thread.
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Timestamp Ordering Guarantee

HTTP 201 Created returned at 09:15:20.104Z; Handler logged at 09:15:21.618Z.
Lag: +1514ms (Verified API responds before background work)

Error Isolation Listener

taskEmitter.on('error') isolates handler exceptions from API crashes.
Fault Tolerance: High (No unhandledRejection)

Github Link : <https://github.com/Ranakeyur-31525/awdf>